

MANAGEMENT REVIEW

Management without theatre.

KPIS

OTIF AND THE PERFECT ORDER

Why the median company ships one order in ten with a failure, why 99% on every part gives 96%, a retailer's rising bar, which date makes a delivery on time, and a card for the perfect order.



1 in 10

orders shipped by the median company has some failure or defect, in APQC's benchmarking.

APQC, 2018

INSIDE

COVER STORY

One order in ten

PAGE 03

THE MODEL

Which date, which unit

PAGE 05

TOOL OF THE ISSUE

The perfect-order card

PAGE 08

NO. 27 · KPIS

In this issue

Logistics reports on time and in full as if everyone meant the same thing. This issue asks what makes an order perfect, who decides when a delivery is on time, and why good rates can still leave one order in ten short.

In the SCOR model an order is perfect only if it arrives in full, on time to the date first promised, undamaged and with correct documents. In APQC's benchmarking the median company scores 90, and 99% on each of the four parts gives only 96%. Walmart raised its bar for full truckloads from 75% to 87% between 2017 and 2019. McKinsey found no standard definition of OTIF in the consumer sector, and a case study found that missing shared definitions held customers and suppliers back.

Stiven Janaqi EDITOR · STIVEN CATALYST

03 COVER STORY
One order in ten

04 THE NUMBERS
A bar that kept rising

05 THE MODEL
Which date, which unit

06 WHAT THE RESEARCH SAYS
When the definition is not shared

07 HOW IT IS MEASURED
Counting perfect orders

08 TOOL OF THE ISSUE
The perfect-order card

09 SOURCES
Sources and method

HOW TO READ THIS ISSUE

Figure

Every figure has its source and year at the foot of its page.

Our reading

Where the editors interpret rather than the research, it says so.

Practice

The steps and the card are proposals to try, not research results.

COVER STORY

One order in ten

APQC compares the perfect order across companies. At the median company the index is 90: one order in ten that it ships has some failure or defect. The top quartile reaches 95 or more.

APQC'S WORKED EXAMPLE: FOUR RATES AND THE SHARE OF PERFECT ORDERS THEY GIVE (%)

98	97	99	82	77.2
On time	In full	Undamaged	Documents	Perfect

The four rates are multiplied, so the perfect order can never be higher than the weakest of them. APQC notes that a company with 99% on each of the four parts reaches only 96% overall.

OUR READING

Four rates that each look good can still leave one order in ten short. The customer receives the order, not the rates.

Sources: APQC, IndustryWeek, 2018; APQC, Open Standards Benchmarking, 2026

The example's rates are illustrative, not a company's data: $0.98 \times 0.97 \times 0.99 \times 0.82 = 0.772$. The median and the top quartile were published in 2018; the sample is not given.

THE NUMBERS

A bar that kept rising

From August 2017 Walmart fined US suppliers whose deliveries did not arrive on time and in full: 3% of the cost of the goods affected. Arriving early counted as a failure too.

WALMART'S ON-TIME, IN-FULL THRESHOLD FOR FULL TRUCKLOADS



A target of 95% within one day had been announced for February 2018. In January 2018 Walmart set it aside: from April the bar was 85% for full truckloads and 50% for smaller loads, up from 33%, within a two-day window. In March 2019 it rose to 87%, and Walmart said it would measure on time and in full separately, so that suppliers would focus on completeness.

OUR READING

A threshold says what the customer will fine, not how well the supplier delivers. And it moves when the customer decides.

Sources: Talk Business & Politics, 2018; Supply Chain Dive, 2019

Announced rules, not measured results, as reported by the trade press; Walmart's own pages were not consulted. Some reports call the two groups large and small suppliers.

THE MODEL

Which date, which unit

OTIF counts the deliveries that reach their destination in the quantity and at the time written on the order. In 2019 McKinsey found no standard definition of either half in the consumer sector.

On time: by which date?

the date the retailer requests

the date the manufacturer promises

a set slot or an agreed window

In full: counted how?

by order

by line

by case

SURVEY OF 24 RETAILERS AND MANUFACTURERS, NORTH AMERICA

An industry standard for OTIF would create value

92%

Prefer to count in full by case

79%

Prefer the requested delivery date

67%

McKinsey proposed counting the cases delivered by the requested date as a share of those ordered, one day early accepted; the survey found no agreement on the window.

OUR READING

Two companies can both report 95% and measure different things. Before comparing OTIF, ask which date and which unit.

Source: McKinsey & Company, 2019

Preferences of 24 large companies, not measured performance; the date of the survey is not given. The definition is McKinsey's proposal, not an adopted standard.

[More in the essay](#)

What last mile taught me about running real operations

WHAT THE RESEARCH SAYS

When the definition is not shared

Helena Forslund and Patrik Jonsson followed six customer–supplier pairs in manufacturing. Missing shared definitions of the metrics, and gaps in the ERP systems, were the main obstacles to managing delivery performance together; the practical problems showed above all in measuring on-time delivery.

In 2022 a McKinsey survey of 35 senior executives at 28 consumer-goods companies in North America found the definitions still moving. More than half said retailers had tightened their OTIF rules, with narrower windows and higher fines.

85%

said at least one key retailer had moved, in the past 12 months, from the requested delivery date to a planned or promised one

17%

said they recover more than 75% of their real cost to serve

OUR READING

When the customer changes the definition, the same work gets a different score. Agree on the definition before arguing about the number.

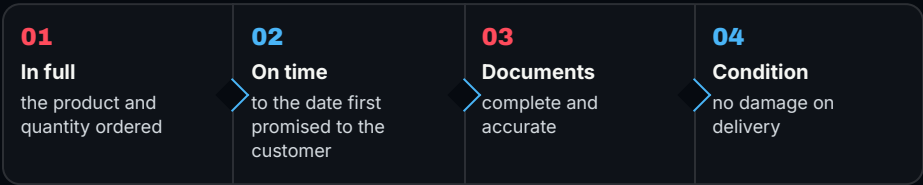
Sources: Helena Forslund & Patrik Jonsson, *International Journal of Physical Distribution & Logistics Management*, 2007; Shruti Lal & Colin Regnier, *McKinsey & Company*, 2022

A case study of six pairs, not a statistical sample; the survey figures are self-reports of 28 companies.

HOW IT IS MEASURED

Counting perfect orders

In the SCOR model, perfect order fulfillment (RL.1.1) is perfect orders divided by all orders. An order is perfect only when every line in it is perfect: one error in one line takes the whole order out.



HYPOTHETICAL EXAMPLE, ONE MONTH

Orders	200: 6 late, 4 incomplete, 2 damaged, 3 with wrong documents
Counted	the 15 failures fall on 12 orders: 188 of 200 = 94.0%
Multiplied	$97\% \times 98\% \times 99\% \times 98.5\% = 92.7\%$

SCOR counts orders, APQC multiplies rates: from the same orders they give different figures, so say which one you report. The orders are invented.

Sources: ASCM, 2026; APQC, Open Standards Benchmarking, 2026

SCOR times the delivery against the date first promised to the customer, not the date requested; the cost of the order is measured elsewhere in the model.

Open the tool: [Delay Analyzer](#)

stivencatalyst.com/tools/delay-analyzer/

TOOL OF THE ISSUE

The perfect-order card

Check a sample of orders each week against the four conditions, with the definitions agreed with the customer written at the top. Count the reasons, not only the rate.

01 Definitions on time to which date and window; in full by order, line or case

02 Order number, customer, number of lines

03 In full every line in the quantity ordered: yes or no

04 On time against the agreed date and window: yes or no

05 Condition and documents any damage; anything wrong or missing in the papers

06 Perfect or not if not, the first condition that failed and its cause

Source: ASCM, 2026

A practice proposed by the editors, after the SCOR model (RL.1.1).

[Open the tool: Last-Mile CX Control Tower](#)

stivencatalyst.com/tools/cx-control-tower/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

**SCOR Digital Standard, Reliability: RL.1.1
(perfect order fulfillment)**

ASCM, 2026

<https://scor.ascm.org/performance/reliability/RL.1.1>

Perfect order performance

APQC, Open Standards Benchmarking, 2026

<https://www.apqc.org/resources/benchmarking/open-standards-benchmarking/measures/perfect-order-performance>

**Achieving the Impossible Dream: Perfect
Order Performance**

APQC, IndustryWeek, 2018

<https://www.industryweek.com/supply-chain/article/22026774/achieving-the-impossible-dream-perfect-order-performance>

**Wal-Mart updates suppliers on new OTIF
requirements**

Talk Business & Politics, 2018

<https://talkbusiness.net/2018/01/wal-mart-updates-suppliers-on-new-otif-requirements/>

**Walmart tightens on-time, in-full rate for
suppliers to 87%**

Supply Chain Dive, 2019

<https://www.supplychaindive.com/news/walmart-on-time-in-full-87-suppliers/550083/>

**Defining 'on-time, in-full' in the consumer
sector**

McKinsey & Company, 2019

<https://www.mckinsey.com/capabilities/operations/our-insights/defining-on-time-in-full-in-the-consumer-sector>

Great service—but who's paying?

Shruti Lal & Colin Regnier, McKinsey & Company, 2022

<https://www.mckinsey.com/capabilities/operations/our-insights/great-service-but-whos-paying>

**Dyadic integration of the performance
management process: A delivery service
case study**

Helena Forslund & Patrik Jonsson,
International Journal of Physical Distribution &
Logistics Management, 2007

<https://doi.org/10.1108/096000030710776473>

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

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